

Study Questions: Planning

1. Define **Planning** in your own words, specifically as it applies to High School.
2. How does the Free Energy Principle constrain our understanding of Planning?
3. Contrast the Classical view of Planning with the Active Inference view.
4. In a decision tree, you evaluate the expected value $E[X] = \sum(x_i * p_i)$ at each branch. Calculate the expected value for a decision with outcomes: A (value 90, probability 0.3), B (value 70, probability 0.5), and C (value 40, probability 0.2).
5. Explain how Dijkstra's algorithm finds the shortest path through a weighted graph. How does the priority queue determine which node to explore next?
6. How does Active Inference planning differ from classical planning? What does it mean to minimize "expected free energy" rather than maximize "expected reward"?
7. A student has 6 hours to study for two exams. Exam A is worth 40% of their grade and Exam B is worth 60%. Using expected value calculations, how should they allocate study time to minimize total expected grade loss?
8. What is a policy in the context of Active Inference planning? How does it differ from a single action? Represent a simple policy mathematically as a sequence of state-action mappings.
9. Explain the role of the planning horizon in Active Inference. What mathematical tradeoff arises between planning further into the future (more accurate) and computational cost?
10. In Dijkstra's algorithm, each edge has a weight. How would you interpret these weights in an Active Inference planning context? What does the "shortest path" minimize?
11. Compare greedy planning (choosing the locally best action at each step) with optimal planning (evaluating all possible action sequences). Give a mathematical example where greedy planning fails to find the best overall plan.

12. How does uncertainty about future states affect planning? If an agent is unsure whether action A will lead to state S1 (good) or S2 (bad), how does expected free energy account for this uncertainty?
13. Construct a simple decision tree for choosing between three weekend activities. Assign probabilities and values to each outcome. Calculate which activity has the lowest expected free energy (highest expected reward minus epistemic value).
14. What is the mathematical relationship between the depth of a decision tree and the computational complexity of evaluating it? Why must real agents limit their planning depth?
15. Explain how "epistemic actions" (actions taken to gain information) appear in Active Inference planning. Give a mathematical example where an agent should explore an uncertain option rather than exploit a known good option.
16. A robot must navigate from point A to point B in a grid with obstacles. Formulate this as a graph search problem and explain how expected free energy guides the robot's path selection.
17. How does Active Inference handle planning when the generative model is wrong? Describe mathematically what happens when the agent's predicted transition probabilities do not match reality.
18. Compare the expected free energy of two study plans: Plan 1 (study the hardest topic first) and Plan 2 (study the easiest topic first). Under what conditions does each plan have lower expected free energy?
19. What is the connection between dynamic programming (breaking a problem into overlapping subproblems) and Active Inference planning over multiple time steps? How does the Bellman equation relate to free energy minimization?
20. Design a planning problem for a high school student deciding which colleges to apply to. Define the state space (acceptance probabilities, costs, preferences), the action space (apply or not), and the expected free energy objective. Calculate the optimal plan for a simplified scenario with 3 colleges.